

Story-Level Control for Multimodal Storybook Generation in Left-Behind Family Communication

Anonymous Author(s)
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Abstract

In Western China, many parents leave rural hometowns to work in cities in hopes of securing better educational and material opportunities for their children. This migration has created many left-behind families, in which parents and children remain emotionally connected but physically apart, with daily contact often reduced to brief, task-focused calls that make children’s feelings and experiences difficult to share across distance. We address this gap with a control-centric multimodal storybook framework that converts child-authored text, drawings, and speech transcripts into plan-constrained picture books for parent sharing. The contribution is an inference-time pipeline with slot-based cue reconciliation, story-level planning, and constraint-based page validation rather than a new trainable backbone. We evaluate it through a controlled 2×2 study on 36 shared source entries and an eight-week observational deployment comparison. In the controlled study, pooled mixed-effects analysis identifies structuring, not multimodal input alone, as the only reliable main effect. A same-text qualitative comparison further shows that structured planning preserves full-day progression and recurring-role identity better than direct text-only generation. In the deployment comparison, the structured cohort trends higher on all four communication indicators, with emotional responsiveness emerging as the clearest reliable difference. Archived psychosocial scales and interview summaries provide exploratory context consistent with structured multimodal storytelling acting as a communication-support scaffold under distance-separated parenting.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**; *Collaborative and social computing*.

Keywords

Multimedia systems, Multimodal storytelling, Narrative planning, Remote family communication, Digital picture books, Left-behind families, Social computing

1 Introduction

Labor-driven migration has produced long-term parent-child separation in many left-behind families in Western China. Although messaging and voice calls are common, many exchanges remain short and task-centric, leaving limited space for children to externalize feelings and for parents to interpret them in time.

Prior work in multimedia and HCI suggests that richer media and narrative artifacts can help children communicate beyond plain text. Yet current systems still fuse sparse multimodal evidence loosely and often lack global narrative control, so outputs may look fluent locally while remaining weak as family-facing emotional signals.

We address this gap through story-level control in a structured multimodal storybook framework for remote family communication. Rather than proposing a new trainable backbone, we introduce an inference-time control layer with slot-based cue reconciliation, story-level planning over emotion/event/role state, and constraint-based page validation. We therefore study two questions: H1, whether multimodal input changes cue grounding and communication-oriented story quality relative to text-only generation; and H2, whether structured planning contributes broader gains in coherence and deployment-level communication indicators.

We study these questions in two phases. A controlled 2×2 factorial study isolates the effects of multimodality and structuring through shared source entries and blind ratings. A complementary eight-week observational deployment comparison tracks communication indicators under sustained use. Figure 1 summarizes the problem, method, and communication value at a glance, and Figure 2 later shows the end-to-end framework on a concrete child-authored example.

Contributions.

- We formulate child-authored storybook generation as a communication-support problem for left-behind families, aimed at externalizing interpretable affective signals for parent follow-up.
- We present a control-centric multimodal framework built around slot-based cue reconciliation, story-level planning, and constraint-based storybook assembly.
- We establish a factorized evaluation lens that links a shared-entry 2×2 controlled study with a real-world deployment comparison, separating modality access and story-level control while offering design and evaluation reference points for future child-centered storybook software.

2 Related Work

2.1 Remote Emotional Communication in Left-Behind Family Contexts

Long-distance family interaction has been widely examined in communication and social computing. Contact frequency alone does not guarantee emotional understanding or better psychosocial adjustment, and migration-affected families

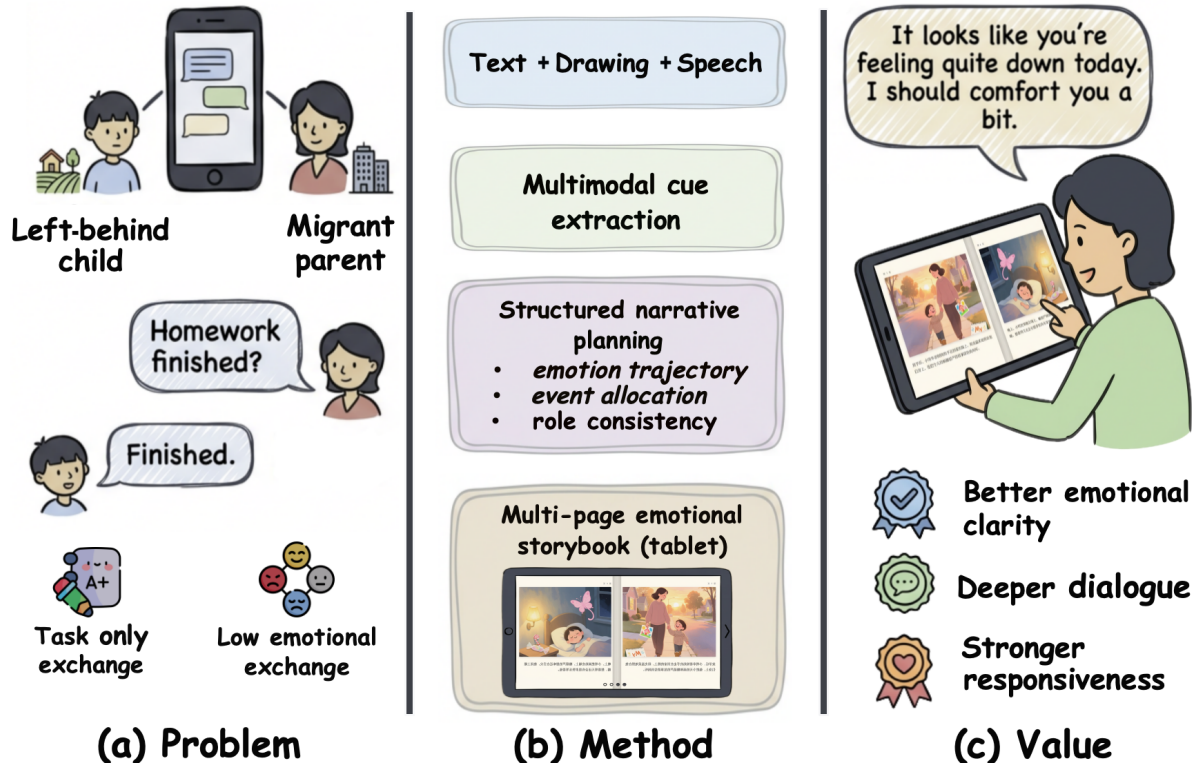


Figure 1: Problem-method-value overview. Left-behind family exchanges are often task-only and emotionally thin; our framework turns child-authored text, drawing, and speech into a planned storybook artifact that supports clearer emotional understanding, deeper dialogue, and stronger parent responsiveness.

often communicate through short coordination-oriented exchanges with affective content left implicit [3, 21, 28]. This pattern is especially salient for left-behind children, who may have limited synchronous time with parents and limited opportunity for reflective disclosure [8, 23, 24]. Family communication systems further show that remote contact is often sustained through shared media spaces rather than voice calls alone [10, 17, 26]. For our task, the bottleneck is therefore not only channel access, but the availability of artifacts that make sparse child-authored emotion legible to remote parents.

2.2 Child-Centered Multimodal Storytelling Systems

Digital storytelling systems for children began with interactive storytelling environments and collaborative authoring tools [2, 18]. Their main contribution is expressive accessibility: they lower dependence on fluent writing and let children externalize experiences through drawing, narration, and co-creation [4, 7, 27]. More recent multimedia systems turn child-authored input into richer visual outputs and stronger short-term engagement [7, 27], but in our setting a storybook can still fail as a remote communication artifact if cues are sparse, ambiguous, or inconsistent across pages.

2.3 Structured Narrative Planning in Generative Pipelines

Structured planning is a well-established way to improve long-form coherence by introducing sequence-level control before local decoding [6, 15, 25]. In visual story generation, representative systems optimize cross-image consistency for pre-written storylines [1, 11–14, 19], and VinaBench shows that source faithfulness and cross-frame consistency should be evaluated jointly [9]. Our task differs because the input is sparse child-authored multimodal expression rather than a complete story prompt, and the outputs must function as family communication artifacts rather than only as offline benchmark generations. Factorial evidence on the separate and interactive effects of multimodal cue integration and narrative structuring remains limited [5, 22]; this is the gap our paper targets.

3 Method

3.1 Framework Overview

Given child-authored multimodal input $\mathbf{x} = \{x^t, x^d, x^s\}$, where x^t denotes text input, x^d drawing input, and x^s speech input transcribed into text, our goal is to generate a multi-page storybook $\mathbf{y} = \{y_1, \dots, y_P\}$ that preserves emotional

meaning and supports remote family communication. Instead of treating this as a single long prompt, we expose an intermediate control state

$$\mathbf{z} = \{e, \alpha, \pi, r, \tau, c\},$$

where e is dominant emotion, α emotion intensity, π affective valence, r the recurring-role set, τ topic/event content, and c scene context. The notation is meant to make explicit what the controller must preserve across pages. We summarize the inference-time target as

$$\mathbf{y}^* = \arg \max_{\mathbf{y}} \mathcal{Q}_{\text{page}}(\mathbf{y}, \mathbf{x}) + \lambda \mathcal{Q}_{\text{global}}(\mathbf{y}, \mathbf{z}),$$

where $\mathcal{Q}_{\text{page}}$ captures page-level expressiveness and relevance, $\mathcal{Q}_{\text{global}}$ captures emotional continuity and role consistency, and $\lambda > 0$ balances local quality against global coherence. This is not a trainable loss; it is shorthand for a system that plans globally and renders locally under explicit constraints.

The framework has four stages: multimodal cue extraction, structured narrative planning, page-level generation, and quality-aware output assembly. The core design choice is to decouple global planning from local rendering: we first build a story-level plan, then render pages under that plan. Figure 2 provides the end-to-end walkthrough referenced throughout the paper.

The pipeline differs from plain multimodal prompting in three ways. First, all modalities are projected into a shared slot schema, so conflicts between event facts, emotional cues, role mentions, and visual anchors are exposed before generation. Second, the planner outputs a page-control table rather than a free-form outline. Third, assembly validates trajectory and role constraints before release, so the final stage is an active controller rather than a simple concatenation step.

3.2 Story-Level Control

Cue extraction and reconciliation. We do not train a dedicated multimodal encoder. Instead, each modality is converted into structured cues by fixed extraction modules and lightweight normalization. Text contributes autobiographical facts, roles, settings, and emotional descriptions; drawings contribute visual anchors; and speech contributes event and affect additions. We denote the extracted cue sets as

$$\mathbf{q}^t = E_t x^t, \quad \mathbf{q}^d = E_d x^d, \quad \mathbf{q}^s = E_s x^s,$$

where E_t, E_d, E_s are prompt-based extraction modules rather than trainable encoders.

The system then merges these cues into a single structured state through rule-guided reconciliation:

$$\mathbf{z} = R\mathbf{q}^t, \mathbf{q}^d, \mathbf{q}^s = \{e, \alpha, \pi, r, \tau, c\},$$

where text anchors autobiographical facts, speech adds non-redundant event or affect details, drawings contribute object and scene anchors, and conflicting role mentions are canonicalized into a recurring-character table with uncertainty flags. In other words, multimodal fusion is handled by an explicit controller state rather than by end-to-end learned fusion weights.

Narrative planning. Given $\mathbf{z}$, we construct a story-level plan

$$\mathcal{P} = \{\gamma_{1:P}, \rho_{1:P}, \kappa\},$$

where $\gamma_{1:P}$ denotes the page-wise emotion trajectory, $\rho_{1:P}$ the page allocation of story events, and κ the cross-page role-consistency constraints.

For the six-page setting used in our experiments, the planner distributes source facts and target emotions across setup, development, peak, and closure while preserving recurring roles and key objects. This plan prevents page drafts that are locally fluent but globally inconsistent.

Structured versus unstructured modes. In unstructured generation, each page is produced directly from local multimodal cues without an explicit story-level plan. In structured generation, each page p is conditioned on both local cues and plan state:

$$y_p \sim p y_p \mid \mathbf{z}, \gamma_p, \rho_p, \kappa.$$

Here y_p denotes the caption-image output for page p , γ_p the target affective state for that page, ρ_p the page-specific event allocation, and κ the shared role-consistency constraint. Operationally, this means the page caption and image prompt are grounded by page-specific source facts, target emotion intensity, recurring-role constraints, and continuity checks. The structured condition therefore changes the control signal and normalization logic rather than backbone model or page budget.

3.3 Rendering and Communication Targets

For each page, the system produces a story caption and an image-generation prompt. Prompt construction injects scene descriptions, recurring-role anchors, and visual constraints for cross-page consistency. After drafting, the controller checks fact coverage, compatibility with the target trajectory, recurring-role consistency, and contradictions between source cues and rendered content. Failed pages are rerendered under tighter constraints up to a small fixed retry budget.

To connect generation quality with communication value in real families, we track four deployment indicators: parent response rate, response length, dialogue depth, and emotional responsiveness ratio. These respectively capture whether a child entry receives a parent reply, how long those replies are, how many alternating turns follow, and how often a parent reply explicitly acknowledges, labels, validates, or follows up on the child’s affective state rather than only events or logistics. The first three are count-like behavioral aggregates, whereas emotional responsiveness is annotation-derived and therefore interpreted more cautiously. In the main text, we report family-level aggregates together with descriptive week-level traces.

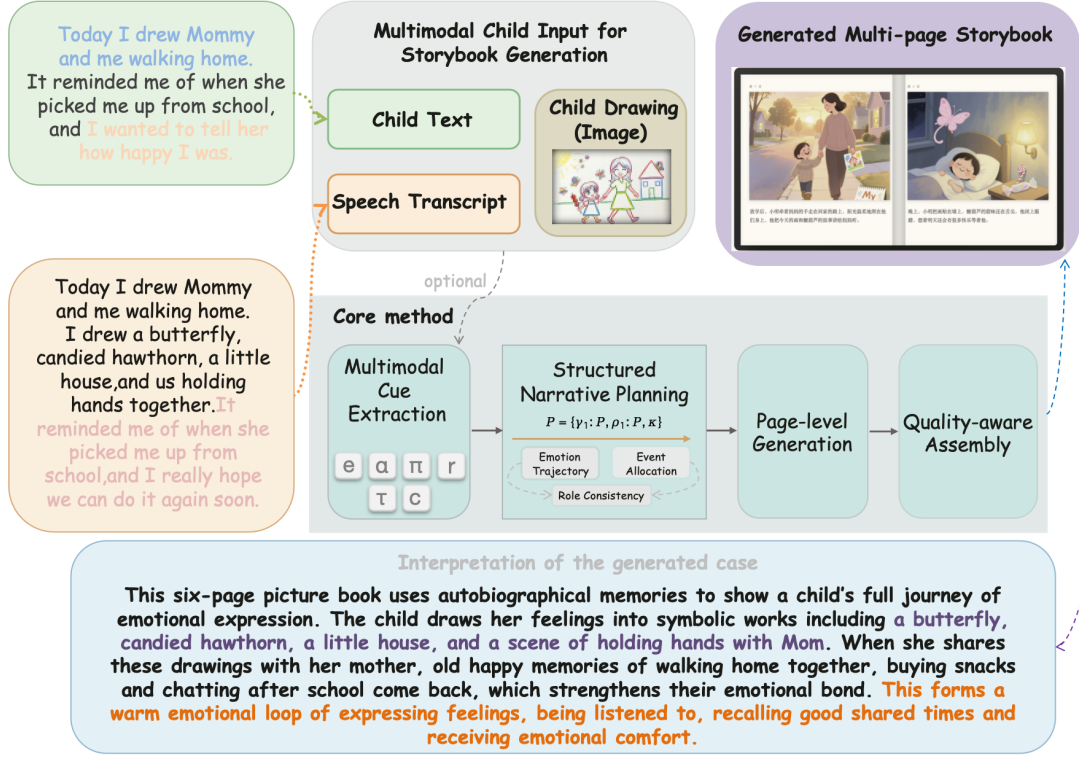


Figure 2: End-to-end walkthrough of the framework on a concrete child-authored example. Child text, speech, and drawing cues are fused into a structured state, expanded into a story-level plan, and rendered into a parent-facing multi-page storybook.

4 Experimental Setup

4.1 Data and Evaluation Protocol

We follow a two-stage evaluation design: controlled mechanism analysis and deployment-oriented communication analysis.

Controlled dataset. We construct a controlled set of child-authored multimodal entries from de-identified study records and analysis exports, each containing short text, one drawing, and one speech transcript. The controlled pool contains 36 source entries, and each source entry is rendered once under each condition of the 2×2 factorial design, yielding $N_{\text{ctrl}} = 144$ storybooks in total (36 per condition): TEXT, TEXT+STRUCT, MM-NoSTRUCT, and FULL. This shared-entry design supports matched across-condition comparison at the source-entry level while keeping controlled samples disjoint from field outcome samples at the family level.

Field dataset. We analyze an eight-week field deployment involving $N_{\text{fam}} = 92$ participating families (46 structured, 46 non-structured) from the same regional context in Western China. The protocol was reviewed under the authors' institutional ethics process; identifiers are omitted for anonymous review. Adult parents provided consent, minors participated with legal-guardian consent and age-appropriate assent. The reported tables are derived from the communication traces of these participating families together with weekly blinded

annotations of sampled exchanges. The current archive preserves equal-sized structured and non-structured cohorts, but it does not preserve a randomized assignment log, a pair-matching sheet, or a recruitment-flow ledger that would let us reconstruct why the archived split is exactly 46 vs. 46. We therefore interpret the between-cohort deployment results as an observational comparison within a real field study rather than as a randomized field experiment, and we treat the preserved balance as an archive-level cohort property rather than as evidence of assignment control. Because the study involves minors and private family communication, participant-level records and provenance tables are retained for internal audit only and are not publicly released. At the preserved archive-summary level, the enrolled and analyzed cohorts are reported as identical, with no preserved record of post-hoc exclusions.

Exploratory triangulation subset. The archive also preserves a smaller $N = 25$ intervention subset with archived EESC [16], CDI [20], child well-being, interviews, and 200 picture-book diary entries. Because this subset is not aligned with either the factorial comparison or the 92-family deployment comparison, we use it only as exploratory context rather than as a primary endpoint.

Blind rating protocol. Controlled outputs are anonymized and rated at the complete storybook level by $N_{\text{rater}} = 24$ independent raters; each six-page sample is scored by $k = 4$ raters.

We use four 7-point quality dimensions: *Emotional Consistency*, *Expressive Richness*, *Visual Relevance*, and *Overall Quality*. These capture cross-page affective coherence, narrative/emotional detail, alignment between generated visuals and input cues, and overall utility as a communication-support artifact. Rater assignment is balanced across conditions, and raters do not see condition labels, family identifiers, or deployment metadata.

For field outcomes, we aggregate the four communication indicators at family level and also inspect their week-level means across the eight-week deployment.

4.2 Conditions, Fairness Controls, and Implementation

The four factorial conditions form our internal factorial baselines:

- TEXT: text-only input, no global narrative planning.
- TEXT+STRUCT: text-only input with global planning.
- MM-NOSTRUCT: text+drawing+speech input, no global planning.
- FULL: text+drawing+speech input with global planning.

Directly comparable public baselines are limited in our setting. Recent visual-storytelling systems typically assume fully written story prompts, benchmark-style story continuations, or evaluation setups that do not match child-authored diary text+drawing+speech inputs, parent-facing communication artifacts, or our six-page diary-to-storybook protocol. Rather than force an unfair cross-protocol benchmark, we treat those systems as related-work references rather than direct empirical baselines, and we keep the paper’s main confirmatory evidence inside the controlled 2×2 design.

We also include one same-text qualitative comparison as an illustrative reference, because the mean-score tables cannot directly show whether a storybook preserves full-day progression and recurring roles across pages. Across the internal factorial conditions, generation budget and backbone are fixed while modality availability and structured-planning switches are varied.

The framework is implemented in Python as a modular orchestration layer over fixed text-understanding, planning, and image-rendering components. Generation uses six pages per storybook at 768×768 resolution, with the same frozen component versions, page budget, and page-level retry ceiling across all internal conditions.

Structured mode activates emotion-trajectory planning, narrative role allocation, and cross-page role-consistency constraints; unstructured mode uses the same page budget but disables story-level planning and normalization. Because validation and rerender checks are part of the control layer itself, the structured condition should be read as the full structured pipeline rather than as a pure planning-only intervention under identical correction dynamics. We do not separately equalize or analyze retry-trigger rates across conditions in the anonymous summary package; any difference in validator-triggered rerender frequency is therefore treated as part of

the pipeline behavior rather than isolated as an independent mechanism. The deployed system is a lightweight dual-portal research prototype in which child and parent interfaces share the same backend orchestrator and the same entry, job, artifact, and notification records.

4.3 Statistical Analysis

Because ratings are nested by both samples and raters, we use linear mixed-effects models rather than ANOVA. For the pooled analysis across the four quality dimensions:

$$\begin{aligned} \text{Score}_{ijm} = & \beta_0 + \beta_1 \text{MM}_i + \beta_2 \text{Struct}_i \\ & + \beta_3 \text{MM} \times \text{Struct}_i + \delta_m \\ & + u_{si} + v_{rj} + \epsilon_{ijm}, \end{aligned}$$

where δ_m is a fixed effect for rating dimension, and u_{si} and v_{rj} are sample and rater random intercepts.

We use this pooled model to estimate overall factorial effects, while the metric-specific mean differences in Table 1 are treated as descriptive rather than four separate confirmatory tests. For the family-level deployment aggregates, we additionally compute two-sided Mann–Whitney U tests with Holm correction across the four metrics. Because the field component is observational rather than randomized, we interpret these checks as between-cohort association tests rather than causal estimators. Week-level traces remain descriptive, same-text qualitative comparison remains illustrative only, and the module-level ablation contrasts are interpreted as within-entry directional evidence under a modest controlled pool rather than as precise population effect estimates.

5 Results and Analysis

5.1 Main Results

The central empirical result is that story-level structuring, rather than multimodal access alone, remains the most reliable contributor in our setting. Table 2 shows this directly in the pooled mixed-effects model: the multimodal coefficient remains positive but non-significant ($\beta_{\text{MM}} = 0.104$, $p = 0.120$), whereas structuring remains positive and statistically reliable ($\beta_{\text{Struct}} = 0.201$, $p = 0.003$). The interaction term remains negligible ($\beta_{\text{Int}} = -0.017$, $p = 0.855$). Residual clustering is again carried primarily by raters ($\text{ICC}_{\text{rater}} = 0.353$), while the sample random effect reaches a boundary fit, suggesting little residual sample-level variance once the condition terms and rater clustering are accounted for. Fixed effects explain a modest share of variance ($R^2_{\text{marginal}} = 0.116$, $R^2_{\text{conditional}} = 0.428$). We therefore read the pooled model as directional factorial evidence for structuring rather than as a rich decomposition of sample heterogeneity. Accordingly, H2 receives pooled inferential support, whereas H1 remains descriptive and weaker than the structuring effect.

Table 1 and Figure 3 show how this inferential pattern appears in the condition means. FULL achieves the highest mean score on all four dimensions, and the clearest descriptive separation again comes from structured planning: TEXT+STRUCT improves over TEXT on all four metrics and remains close to FULL on Overall Quality (4.701 vs. 4.750). By contrast,

Table 1: Blind-rating results for the controlled 2×2 design (7-point scale, mean $\pm$ SD over sample means).

Condition	Emo. Cons.	Expr. Rich.	Vis. Rel.	Overall
TEXT	4.472 ± 0.540 4.611	4.389 ± 0.512 4.688	3.868 ± 0.721 4.062	4.528 ± 0.553 4.701
TEXT+STRUCT	± 0.516 4.729	± 0.464 4.431	± 0.398 3.951	± 0.353 4.562
MM-NOSTRUCT	± 0.596 4.812	± 0.556 4.708	± 0.719 4.139	± 0.596 4.750
FULL	± 0.584	± 0.569	± 0.575	± 0.486

multimodal input without structuring yields a narrower pattern of gains: MM-NOSTRUCT exceeds TEXT on Emotional Consistency (4.729 vs. 4.472) but remains close on Overall Quality (4.562 vs. 4.528). This pattern suggests that multimodal cues are still helpful but metric-dependent, whereas structured planning remains the more reliable contributor. Multimodal access is most visible on Visual Relevance, where drawing- and speech-derived cues add anchors that text alone often under-specifies.

Because directly comparable public baselines are not available for child-authored diary text+drawing+speech storybooks in this communication setting, Figure 4 provides a qualitative same-text reference rather than a separate benchmark. Both systems in the figure receive the same de-identified diary text, while only the story-level planner is toggled. The comparison shows that direct text-only generation tends to repeat a few salient local scenes, whereas structured generation covers the diary chronologically and keeps recurring roles more legible across pages. We therefore read this figure as interpretive support for the same design goal reflected in Table 1, not as standalone confirmatory evidence.

Table 3 then connects these quality results to deployment behavior. The structured cohort is higher on all four communication indicators. Secondary Mann-Whitney tests reported in the appendix identify emotional responsiveness as the clearest deployment difference ($U = 1436.5$, uncorrected $p = 0.003$, Holm-adjusted $p = 0.013$, $r = 0.31$). Dialogue depth is only nominal before correction, while response rate and response length remain non-significant. Figure 5 shows that the same ordering holds across all eight weeks. We therefore treat emotional responsiveness, although annotation-derived and more subjective than the count-like reply metrics, as the strongest observational deployment signal, with the other three indicators best read as aligned trends. Read together with archived psychosocial scales and interview summaries, this pattern is consistent with communication that is easier to continue and emotionally respond to, and more broadly with a warmer parent-child emotional link under long-distance separation, while still falling short of a direct causal measure of relationship change.

5.2 Ablation Study

We next isolate module-level contributions in the structured pipeline (Table 4) by removing one module at a time.

The mean-score pattern remains differentiated: removing emotion-trajectory planning yields the largest degradation in

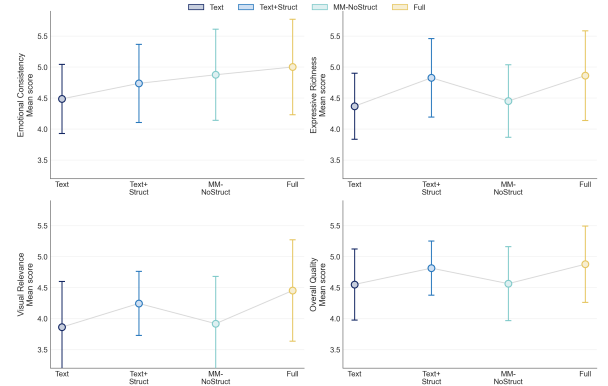


Figure 3: Controlled-study blind-rating means with standard-deviation error bars. Structured planning yields the clearest gains across the four quality dimensions.

Table 2: Pooled linear mixed-effects results for controlled blind ratings.

Term	β	SE	z	95% CI	p
Multimodal main effect	0.104	0.067	1.55	[-0.027,0.236]	0.120
Structured main effect	0.201	0.067	3.00	[0.070,0.333]	0.003
MM $\times$ Struct interaction	-0.017	0.095	-0.18	[-0.203,0.169]	0.855
ICC (rater)				0.353	
ICC (sample)				0.000 (boundary fit)	
R^2 (marginal)				0.116	
R^2 (conditional)				0.428	

Table 3: Family-level deployment outcomes (mean $\pm$ SD over 46 families per group).

Outcome	Structured	Non-structured
Parent response rate	0.583$\pm$0.156	0.559 $\pm$ 0.153
Response length	19.897$\pm$4.525	18.905 $\pm$ 4.041
Dialogue depth	2.872$\pm$0.696	2.563 $\pm$ 0.637
Emotional responsiveness ratio	0.621$\pm$0.092	0.560 $\pm$ 0.103

Emotional Consistency (0.101), removing narrative role allocation yields the largest degradation in Expressive Richness (0.128), and removing the consistency constraint yields the largest degradation in Visual Relevance (0.135). Overall Quality shows its largest drop under the consistency constraint (0.085), followed by role allocation (0.071) and trajectory planning (0.064). Figure 6 visualizes the same module-wise structure, and Table 5 reports paired two-sided t -tests against the full model using matched sample IDs. All planned removals remain significant after Bonferroni correction, with the strongest within-entry standardized separations on Emotional Consistency for trajectory planning ($d = 1.135$), Expressive Richness for role allocation ($d = 1.457$), and Visual Relevance for consistency constraints ($d = 1.718$); given the modest controlled pool, we interpret these values as matched-sample

Shared input cues

late wake-up

math help from classmate

chicken-leg lunch

running race

homework with grandma

Representative diary about a full school day, family care, and a positive closing mood.

Direct text-only generation No global story planner



Fails at character consistency & full-day narrative, only producing fragmented local scenes.

Structured text generation (ours)



Covers the diary chronologically and preserves school, race, and home episodes as a coherent sequence.

Figure 4: Qualitative same-text baseline comparison. Both systems receive the same de-identified diary text, but only the structured condition activates story-level planning. This de-identified case is manually selected as a representative illustration of a full-day diary with multiple distinct episodes and is not used for sampling-based inference. Direct text-only generation repeats a few local scenes, whereas structured generation preserves the diary’s bedroom, classroom, lunch, running, and homework progression as a coherent full-day storybook.

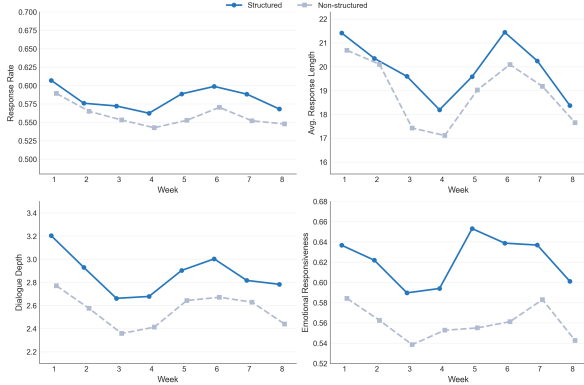


Figure 5: Week-level behavior across the eight-week observational deployment comparison. Each point is the mean over 46 family-level weekly observations per group; the traces are descriptive and shown without uncertainty bands. The structured condition preserves the same ordering advantage over the non-structured condition throughout the observation window.

Table 4: Ablation on structured modules (mean scores).

Variant	Emo.	Expr.	Vis.	Overall
Full model	4.838	4.725	4.163	4.752
w/o trajectory planning	4.737	4.692	4.130	4.688
w/o role allocation	4.798	4.597	4.134	4.681
w/o consistency constraint	4.792	4.666	4.028	4.667

separation magnitudes rather than as precise population effect estimates.

6 Discussion and Limitations

The most defensible claim is that story-level structuring, not multimodal access alone, drives the reliable gain in this setting. The controlled study provides the clearest evidence for that mechanism; the deployment comparison adds one stable observational signal on emotional responsiveness, while the remaining field indicators together with archived psychosocial scales and interview summaries provide convergent supporting evidence that communication becomes easier to continue and easier to respond to in an affectively warmer way. In practical terms, the deployment-side evidence supports the

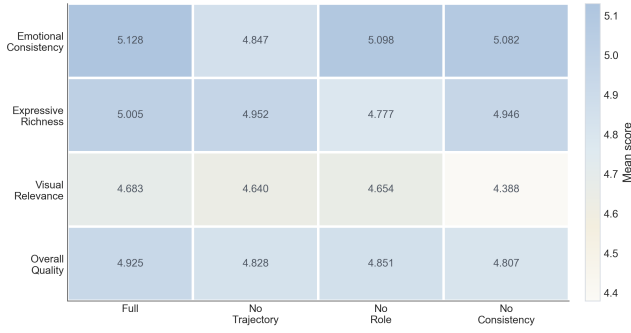


Figure 6: Ablation of the three structured controls, visualized as a metric-by-module mean-score heatmap.

Table 5: Paired ablation contrasts versus the full model (36 matched sample IDs, two-sided; Bonferroni-corrected within metric over three planned removals). Positive mean differences indicate degradation relative to the full model. Cohen’s d is reported as a paired standardized mean difference on the matched sample-level contrasts.

Metric	Contrast (vs. Full)	Mean diff	t	p_{adj}	Cohen’s d
Emo. Cons.	w/o trajectory planning	0.101	6.808	< 0.001	1.135
	w/o role allocation	0.041	3.864	0.001	0.644
	w/o consistency constraint	0.046	5.665	< 0.001	0.944
Expr. Rich.	w/o trajectory planning	0.033	3.123	0.011	0.520
	w/o role allocation	0.128	8.743	< 0.001	1.457
	w/o consistency constraint	0.059	5.557	< 0.001	0.926
Vis. Rel.	w/o trajectory planning	0.033	4.518	< 0.001	0.753
	w/o role allocation	0.029	3.721	0.002	0.620
	w/o consistency constraint	0.135	10.306	< 0.001	1.718
Overall	w/o trajectory planning	0.064	6.135	< 0.001	1.022
	w/o role allocation	0.071	7.331	< 0.001	1.222
	w/o consistency constraint	0.085	8.607	< 0.001	1.434

interpretation that these storybooks can help parents and left-behind children maintain a stronger emotional link across distance, even though the current paper does not isolate that relational effect with randomized causal identification. From a multimedia-systems perspective, the contribution is a control layer that makes sparse child-authored text, drawings, and speech more interpretable under explicit cross-page constraints, including more stable recurring characters across a multi-page storybook. The paper should therefore be read in descending order of evidential strength: controlled factorial gains first, narrower but real field convergence second, and only then broader relational interpretation. At the same time, we do not isolate a causal family-outcome effect from the deployment materials: the eight-week field data are observational rather than randomized, no directly comparable public baseline exists for this task protocol, the control layer includes validation-linked correction dynamics in addition

to planning, and the archived psychosocial materials rely on restricted-access summaries rather than participant-level release. We also do not report seed-sweep, page-budget, or subgroup robustness analyses in the anonymous summary package.

7 Conclusion

We presented a control-centric multimodal storybook framework for remote parent-child communication in left-behind families. Rather than claiming a new generative backbone, the paper contributes an inference-time control layer that reconciles sparse child-authored cues into a story-level plan and enforces cross-page constraints during rendering and assembly, while also offering a practical design reference for future child-centered storybook software.

Taken together, the paper closes the two motivating questions asymmetrically. H1 receives narrower descriptive support: multimodal cues mainly strengthen grounding and visual relevance. H2 receives the clearest inferential support: story-level structuring most reliably improves artifact quality, recurring-role continuity, and the parent readability of the resulting storybook. The same-text qualitative comparison, eight-week deployment, and archived scale-and-interview materials then provide convergent supporting evidence that such artifacts can help sustain warmer and more continuous parent-child communication under separation. The contribution should therefore be read as a communication-support scaffold and evaluation perspective for generated storybook software, with meaningful field-supported promise but without a randomized basis for claiming causal downstream family well-being effects.

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